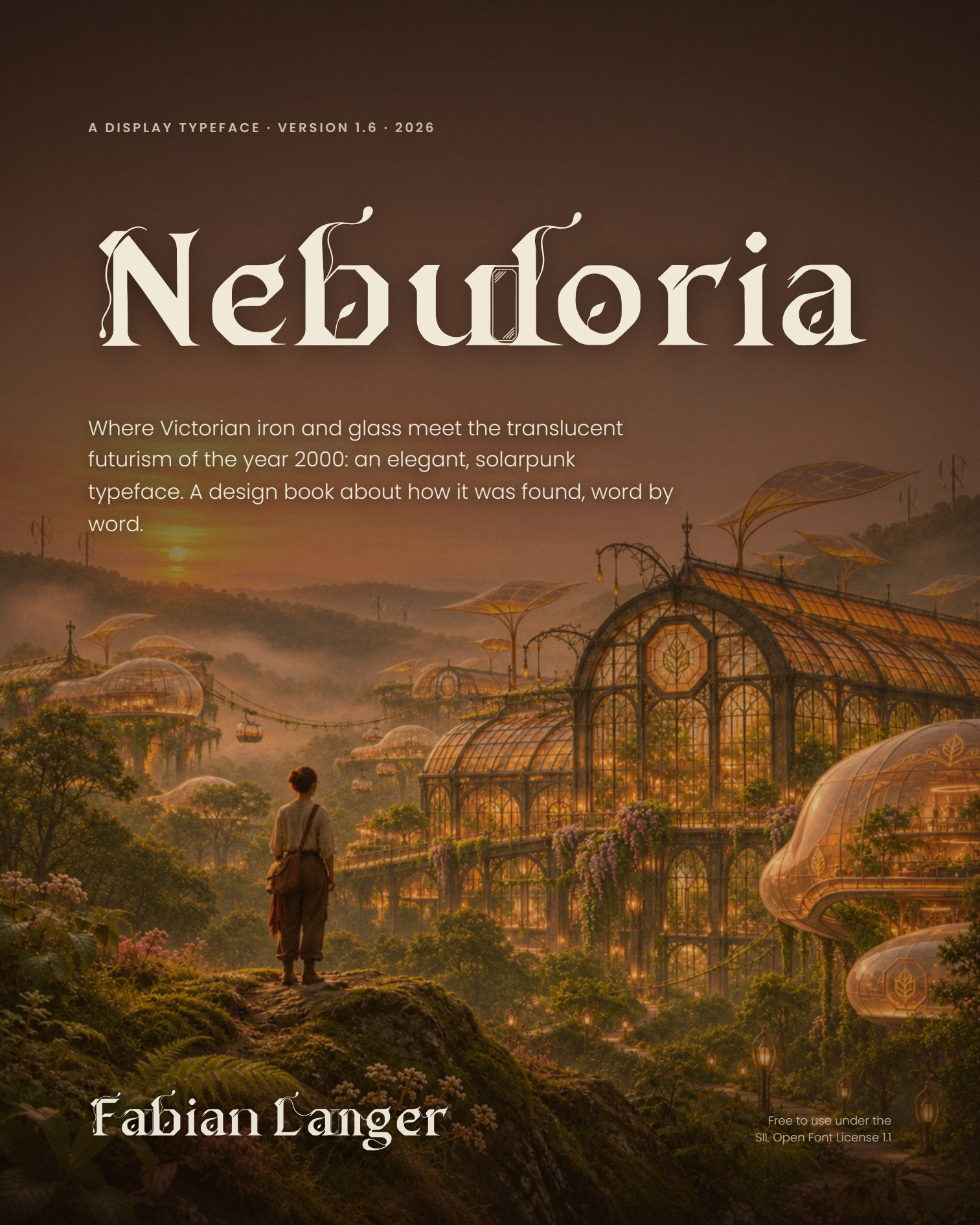
A DISPLAY TYPEFACE · VERSION 1.6 · 2026

# Nebuloria

Where Victorian iron and glass meet the translucent futurism of the year 2000: an elegant, solarpunk typeface. A design book about how it was found, word by word.



Fabian Langer

Free to use under the
SIL Open Font License 1.1

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# Why Nebuloria

Nebuloria began as a signature: a name for a website that needed one human, crafted moment between calm, geometric text.

The brief was a feeling, not a font. Victorian conservatories, where cast iron carries glass and plants climb the columns. And the see-through electronics of the late nineties, where you could look into a telephone and see its wires. And the futurism of that same moment: organic, aerodynamic shapes from aerospace, and the straight lines of microchips. Eras that all made structure visible, and that together describe an optimistic, green and elegant future: refined solarpunk.

The typeface was not drawn by hand. It was *directed*: by learning the language of type design, turning a mood into precise rules, and letting AI models draw and redraw until the rules held. Every principle in this book started as a sentence.

Nebuloria is a display face. It is made for large sizes: wordmarks, headlines, pull quotes. It comes in two styles: **Regular**, calm and readable, and **Hekwerk** (Dutch for "iron fence"), which opens a window between every pair of upright stems.

# Iron and glass




The Crystal Palace, London, 1851. Lithograph by J. McNeven. Public domain.

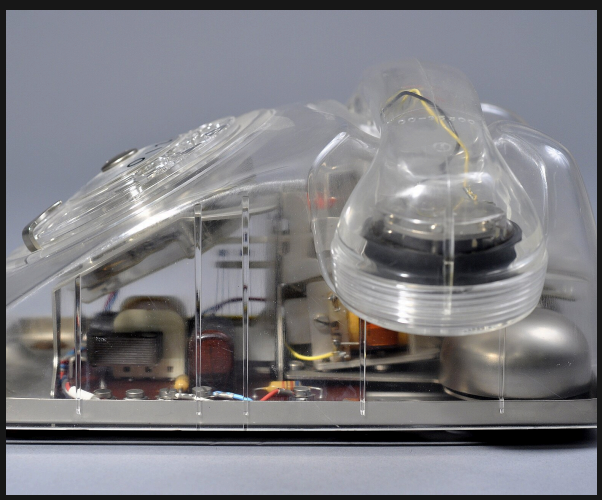


Palm House, Royal Botanic Gardens, Kew. Photo: Daniel Case, CC BY-SA 3.0.

The Victorians built cathedrals for plants: heavy iron ribs carrying hair-thin glazing bars, and between them glass.

From this world Nebuloria takes its **structure**: heavy strokes next to hairlines, scrolls that grow from the line, the octagon and the 45° cut. Not its shine: no gold, no chrome, no metallic effects.

# See-through



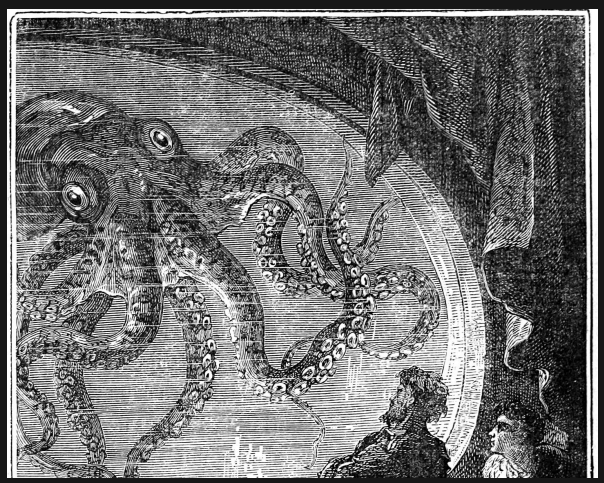
Transparent telephone with visible wiring. Photo: Ellinor Algin / Swedish National Museum of Science and Technology, CC BY 4.0.




iMac G3, translucent Bondi blue, 1998. Photo: katz2110 (Flickr), CC BY 2.0.

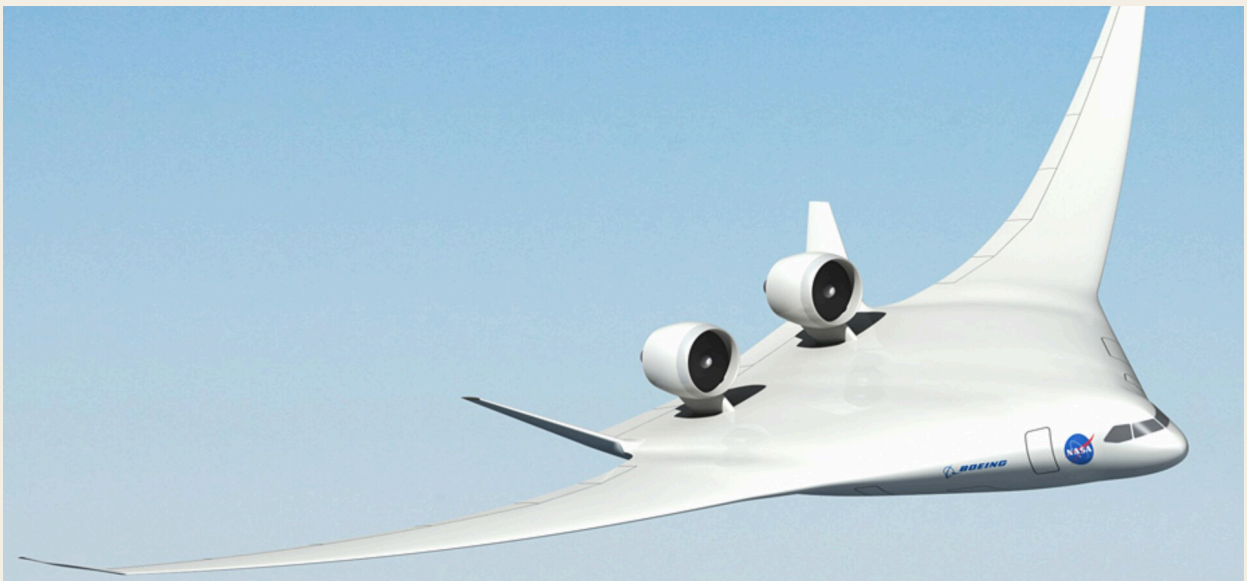
Around the year 2000, electronics turned transparent. The casing became a window, and the circuit inside became decoration.

A font cannot be transparent: it is one colour. So translucency had to become **form**: hairlines, open glass panes, wires that run behind a stroke and show through it.



Fiction got there first: the salon of the Nautilus in Jules Verne's *Twenty Thousand Leagues Under the Seas* (1870), a Victorian future of iron, glass and engineering. Engraving by A. de Neuville and É. Riou. Public domain.

# Future forms

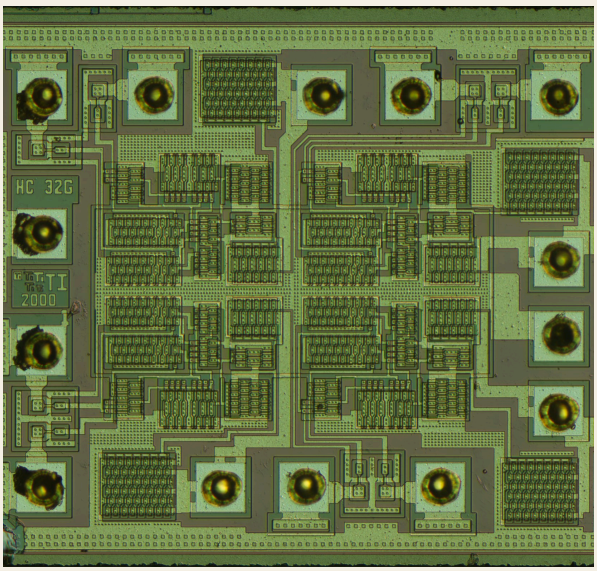



Blended-wing-body aircraft concept. NASA / The Boeing Company, public domain.

Y2K futurism was organic and aerodynamic: one surface that swells, carries its weight and tapers out to a sharp edge.

That is the **weight profile** of every Nebuloria stroke: full and heavy, then running out into a needle point, like a wing into its tip.

Its other half is the microchip: **straight lines**, right angles and 45° turns, packed with precision. From it come the chamfered corners, the windows and the glare lines.




Die of an integrated circuit: straight lines and 45° routing. Photo: Cole L, CC BY-SA 2.0.

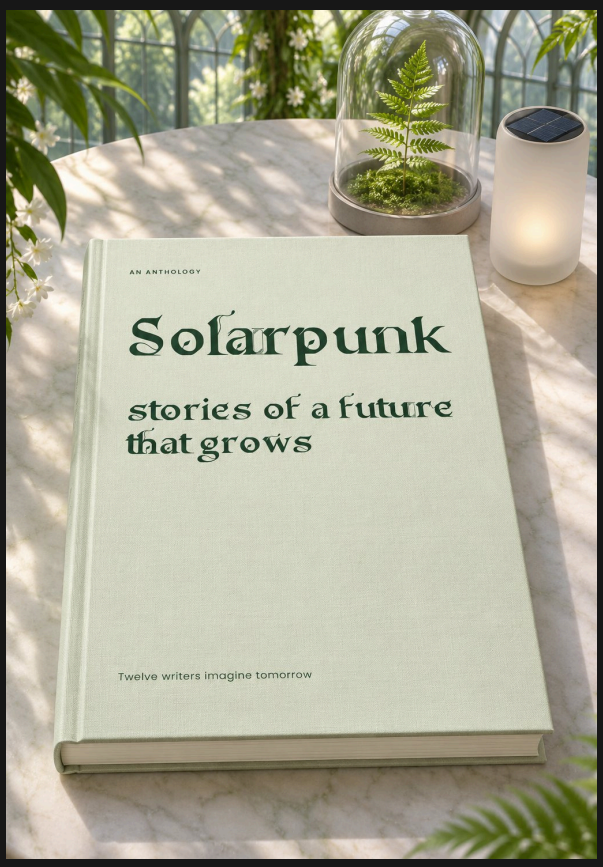
A NAME FOR THE BLEND

# Solarpunk

Put the three together: the Victorian glasshouse, transparent technology and organic engineering. What you get is a future where technology and nature grow into each other.

That future has a name: **solarpunk**. Nebuloria speaks its **elegant** dialect: not makeshift or rustic, but refined, crafted and precise. Quiet luxury in glass, iron and green. Solar glass in an orangery roof, vines on slender frames, cables and roots running side by side.

It is a natural fit for high-end solarpunk worlds: botanical restaurants and hotels, wayfinding for gardens and museums, fine publishing, perfume and tea, games and film titles that imagine a hopeful, beautifully made future.




A linen-bound solarpunk anthology.




The Orangery: wayfinding for a botanical salon under solar glass.

# Heavy and hairline



Broad-edged and pointed nibs. Photo: R. Henrik Nilsson, CC BY 4.0.

Every main stroke behaves as if drawn with a **broad nib held at 45°**: strokes from top-left to bottom-right are heavy, strokes from top-right to bottom-left are hair-thin. It is the same contrast as iron ribs against glazing bars.

| MEASURE | UNITS (OF 1000) |
|---|---|
| Lowercase stem | ≈ 112 |
| Thinnest hairline | ≈ 13 |
| Contrast | ≈ 8 : 1 |



Heavy stems, hairline arches. The thick parts carry the weight, the hairlines carry the elegance.

# The stroke pair



Wrought-iron gate. Photo: MauraWen, CC BY 4.0.



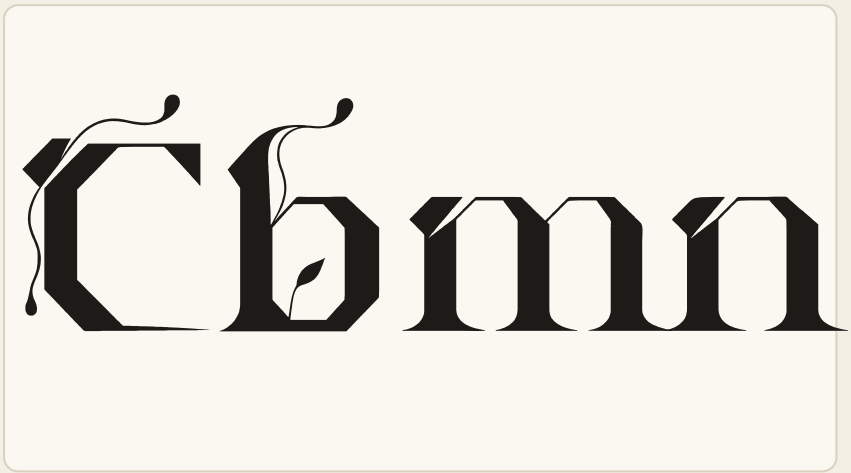
In a wrought-iron gate, thin scrolls grow alongside the heavy bars. In Nebuloria every capital is a **stroke pair**: a heavy stroke and a thin, wavy **tendril** that starts at the same chamfered corner and runs beside it, ending in a small teardrop.

The two have **mirrored weight**: the main stroke is heavy and tapers to a needle point; the tendril is a hairline that swells slightly at its end.

# Forty-five degrees



Octagonal cast-iron bandstand. Photo: Rodhullandemu, CC BY-SA 4.0.



Where a classic letter has a round corner, Nebuloria has a straight cut at 45°.

Bowls and arches are built like the Victorian octagon: straight sides joined by **chamfers**. The cut is where the stroke turns thin. Diagonals and curls only travel one way, from north-east to south-west, the direction of the pen.

This is also where the Y2K world comes in: the chamfer is the bevelled edge of moulded plastic and the 45° route of a circuit trace.

# A leaf in the counter



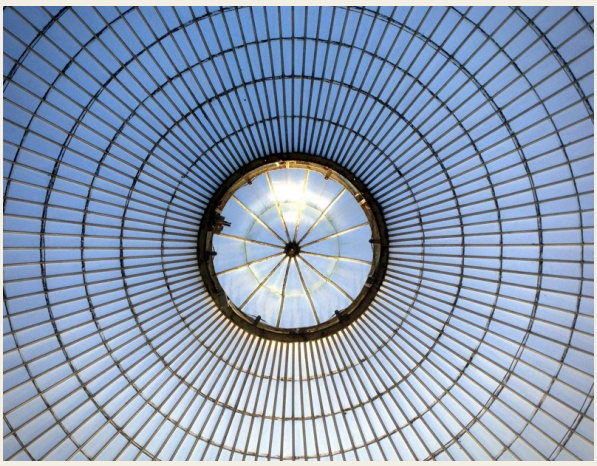
Backlit leaf. Photo: Jon Sullivan, public domain.

A conservatory is a house for plants, so the plants are inside the letters. In every closed bowl (a, b, d, g, o, p, q) one small leaf grows from the inside of the stroke into the counter.

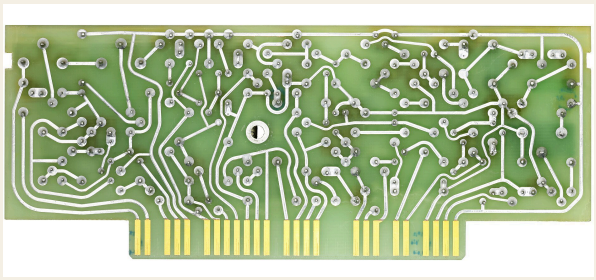
**Botanical, not floral**: the leaf grows from the stroke. Nothing is stuck on top, and there are no flowers.



# The window



Dome of the Kibble Palace, Glasgow. Photo: Lib79, CC BY-SA 4.0.



Printed circuit board with 45° traces. Photo: Mister rf, CC BY-SA 4.0.



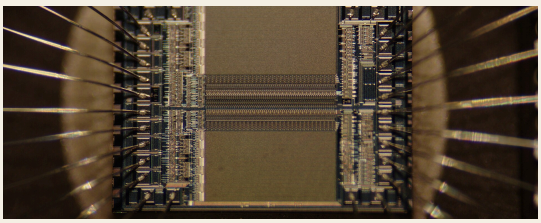

When a letter ending in an upright stem meets a letter starting with one (as in *an*, *il*, *hu*), the two stems are joined by thin bridges and enclose a new counter: a **window**.

Inside, three short 45° lines. They read twice: as the glare on a pane of glass, and as circuit traces seen through clear plastic. One shape, both worlds.

# The wire: cable and vine



Liana in a rainforest. Photo: Dick Culbert, CC BY 2.0.



Bond wires connecting a chip. Photo: Zephyris, CC BY-SA 3.0.



Ribbon cable. Photo: Retired electrician, CC0.

Langer

Between some letters a thin **wire** grows: from the top-right of one letter, hanging in a soft round curve, into the top-left of the next. It is a vine and a cable at once: the liana on the conservatory column and the wire inside the transparent phone. Above all, it **connects**: letters into words, and the two worlds into one.

AND WHAT NEBULORIA IS NOT

# Leave out

- Metallic effects, chrome, gloss, gold. The iron is structure, not shine.
- Flowers or leaves stuck on top of letters. Everything grows from the stroke.
- Fairy-tale "vine lettering". Lianas are rare and precise.
- Art deco and blackletter. The 45° cut is a detail, not a costume.
- Uniform, monolinear strokes. Contrast is the backbone.
- Round, soft blobs. Every end is a needle point, a teardrop or a crisp cut.
- Small sizes. Nebuloria starts at 40 px; its details need 72 px and up.

# Set it free

Nebuloria is free for everyone, for personal and commercial work. Here is how to get it and use it, no technical knowledge needed.

**1 · Download**

Go to **github.com/Fabianlanger/nebuloria-font** and click **Download Nebuloria**. That link always gives you the newest version. Unzip the file (double-click it).

**2 · Install**

**Windows:** open the folder *fonts/otf*, right-click both files and choose **Install**.
**Mac:** double-click each file and click **Install Font**.

Both styles now appear in your font menu: **Nebuloria** and **Nebuloria Hekwerk**.

**3 · Use**

Pick Nebuloria in Word, PowerPoint, Keynote, Pages, Figma, Photoshop or Illustrator. Use it big: titles, names, posters, covers. Set the rest of your text in a calm, simple font.

**Windows and wires**

The glass windows and hanging wires are a hidden extra called **discretionary ligatures**. Switch them on where your app allows it:

| APP | WHERE |
|---|---|
| Word | Font › Advanced › Ligatures: *All* |
| Figma | Type settings › Details › Discretionary ligatures |
| Illustrator, InDesign | OpenType panel › Discretionary Ligatures |
| Photoshop | Character panel › OpenType › Discretionary Ligatures |

For the iron fence, choose the style **Nebuloria Hekwerk**. Apps without these options show the clean letters, which still look great.

**On a website**

Use the files in *fonts/webfonts*. Web developers: see the next page.

Licence: SIL Open Font License 1.1. Use, share, embed and change it freely; just don't sell the font on its own. Letters A–Z and a–z, figures 0–9, punctuation, the euro sign and accented letters (é, ü, ç, ñ …).

# Under the hood

For websites, apps and design tools that read OpenType features directly.

```
@font-face {
  font-family: "Nebuloria";
  src: url("nebuloria-regular.woff2") format("woff2");
  font-display: swap;
}
@font-face {
  font-family: "Nebuloria Hekwerk";
  src: url("nebuloria-hekwerk-regular.woff2") format("woff2");
  font-display: swap;
}

.wordmark { font-family: "Nebuloria", serif; font-feature-settings: "dlig" 1; }
.headline { font-family: "Nebuloria", serif; }              /* extras off */
.fence    { font-family: "Nebuloria", serif; font-feature-settings: "dlig" 1, "ss01" 1; }
```

| FEATURE | NEBULORIA (REGULAR) | NEBULORIA HEKWERK |
|---|---|---|
| `dlig` | Wires + at most one window per word | Wires + a window in every pair of stems |
| `ss01` | A window in every pair | Back to one window per word |

| SIZE | USE |
|---|---|
| 72 px and up | Wordmarks, pull quotes, with `dlig` |
| 40–72 px | Headlines, `dlig` off or sparing |
| below 40 px | Use a text face |

| METRIC | VALUE | METRIC | VALUE |
|---|---|---|---|
| Units per em | 1000 | x-height | 500 |
| Cap height | 700 | Ascender / descender | 904 / -339 |

Windows appear between a letter ending in a stem (a d h i l m n u) and a letter starting with one (b h i k l m n p r u). Wires connect Fa, La, ng, er, re, ce, ge. Source code and build scripts are in the GitHub repository.




Capitals carry the most ornament: every stem a stroke pair, a chamfered flag at the top-left, and an asymmetric foot that runs out into a hairline.




| METRIC | VALUE | METRIC | VALUE |
|---|---|---|---|
| Units per em | 1000 | Cap height | 700 |
| x-height | 500 | Glyphs | A–Z, a–z, 0–9, signs, accents + ligatures |

0 1 2 3 4 5 6 7 8 9

. , : ; ! ? ' " ‚ ’ “ ” . …

- – — ( ) & / + ⅞ # @ €

é è ê ë á à ä â í ì ï

î ó ò ö ô ú ù ü û ç ñ

É È Ê Ë Á À Ä Â Í Ì Ï

Î Ó Ò Ö Ô Ú Ù Ü Û Ç Ñ

Figures are lining: as tall as the capitals, with the same flags, chamfers and leaves. Every dot is the chamfered diamond of the i. Accented letters are not redrawn: each is the font's own letter with one drawn accent on top, so an é is always exactly the e.

# Two styles

Discretionary ligatures off

Fabian Langer
illumination

Nebuloria Regular · `dlig` on: wires, and at most one window per word

Fabian Langer
illumination

Nebuloria Hekwerk · `dlig` on: a window in every pair, the iron fence

Fabian Langer
illumination

Both styles contain both behaviours: `ss01` switches Regular to the fence and Hekwerk back to one window per word. Wires connect *Fa, La, ng, er, re, ce, ge*.

# Fabian Langer

## From strategy to solutions

### Not everything that adapts, learns.

### Confusing concepts make for crippling policy

Designed to be filled with colour and gradients: the hairlines are strong enough for a yellow-to-red gradient at 40 px and above.



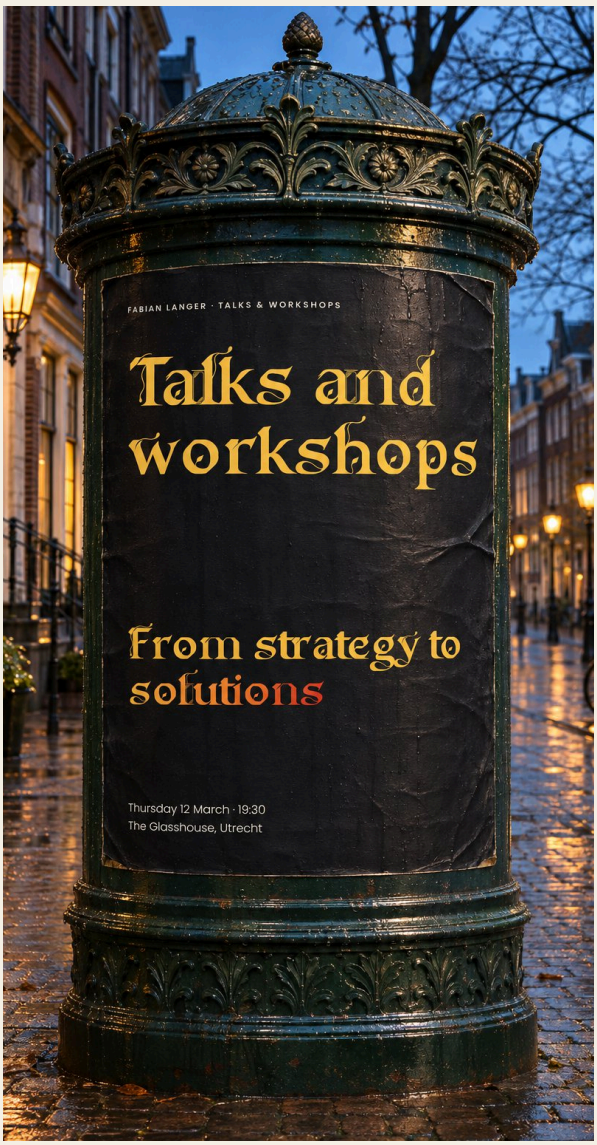

The wordmark in a website hero (role R1), with Poppins for text.



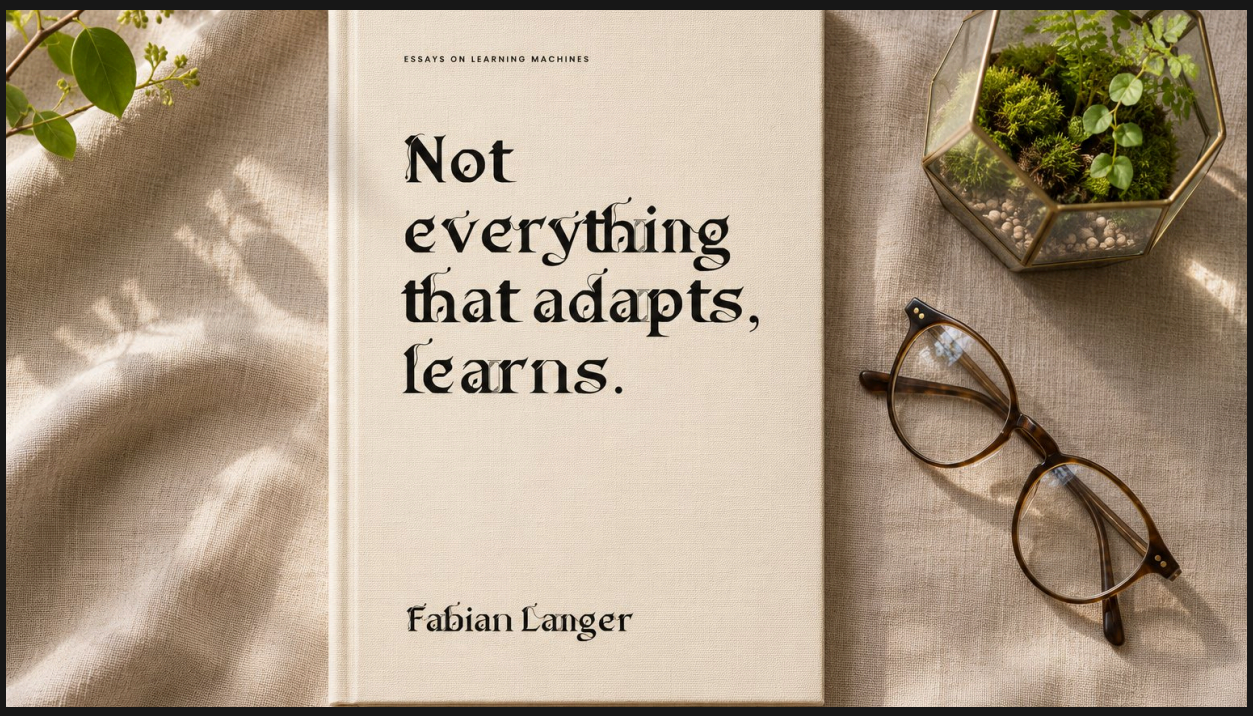

Letterpress business card, warm yellow ink on black cotton paper.




Acid-etched into the door of a glasshouse: the font back in its own world.




Event poster on a cast-iron advertising column (roles R2 and R3).




A cloth-bound essay cover: pull quote as title (role R3).




Nebuloria Hekwerk on a translucent Y2K-style case: the windows meet the real circuit board.

Mockup scenes are AI-generated. The lettering was set in the real font and placed into each scene; small deviations from the font are possible in the generated images.

# From words to letters

Nebuloria was designed by learning the vocabulary of type design and using it to direct AI. Each round followed one loop: **learn a term → write a rule → render with an image model → judge → refine the rule.**



v0.1 · The first F: "stroke pairs", "leaf terminals".




v0.1 · Too many leaves; the model drifts to fairy-tale lettering.




v0.2 · "Chamfers" without balance: everything becomes angular.




v0.2 · Langer: the sharpness that became the standard.



v0.3 · The window ligature, defined for letter classes.




v0.4 · The target image.

The biggest lesson: **words need balance.** "Chamfered corners" without a limit means "everywhere"; "at most one small leaf" means "none". A precise brief says what, how much and where.

# From images to a font



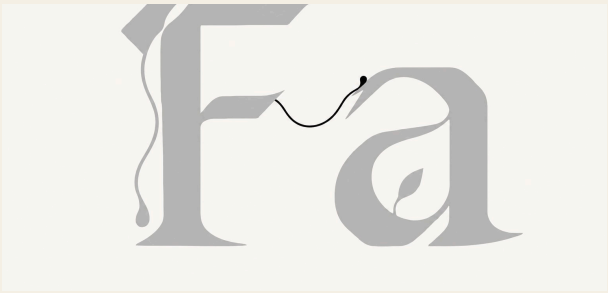

1 · Glyph sheets: one row of letters per image.




2 · Letters that felt too basic were redrawn with stricter rules.




3 · For wires, the real letters were rendered in grey…




4 · …and only the black wire was drawn on top and extracted.

Every letter was cut from its sheet, traced into vector outlines, set on a shared baseline and x-height, and built into an OpenType font with Python (fontTools). Spacing is measured from each letter's heavy body, so tendrils and tails may reach into their neighbours. Windows and wires are OpenType rules, so they appear by themselves while typing.

| VERSION | CHANGE |
|---|---|
| 1.0 | First full alphabet from the glyph sheets |
| 1.1 | C G K M V W X Y and m n u x redrawn with the family's stroke techniques |
| 1.2 | 45° chamfers instead of round curves for C, b, m, n |
| 1.3–1.4 | Windows and wires; calmer windows, round lianas |
| 1.5 | Baseline fixes, final b and F, two styles: Regular and Hekwerk |
| 1.6 | Figures, punctuation, € and accents: drawn on sheets with an H or n as ruler; accented letters built from the font's own letters |

# With thanks

Typeface, art direction and rules: Fabian Langer, 2026. Drawn with AI image models and built with open-source tools (vtracer, fontTools).

## PHOTOGRAPHS AND ILLUSTRATIONS

| IMAGE | AUTHOR | LICENCE |
|---|---|---|
| Crystal Palace - interior.jpg | J. McNeven | Public domain |
| Palms in Palm House atrium.jpg | Daniel Case | CC BY-SA 3.0 |
| Wrought iron gate St Michael's church.jpg | MauraWen | CC BY 4.0 |
| Stanley Park bandstand 2018-1.jpg | Rodhullandemu | CC BY-SA 4.0 |
| Kibble Palace, Botanic Gardens.png | Lib79 | CC BY-SA 4.0 |
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